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Insights into the Competitive Mechanisms of Per- and Polyfluoroalkyl Substances Partition in Liver and Blood

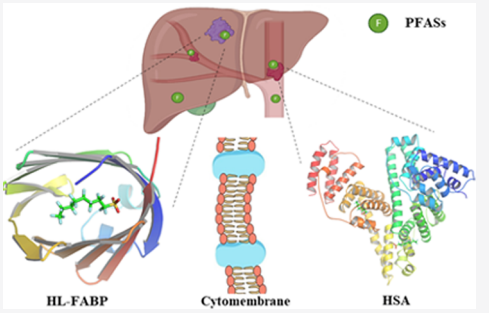
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ABSTRACT: Some per- and polyfluoroalkyl substances (PFASs) tend to be accumulated in liver and cause hepatotoxicity. However, the difficulty to directly measure liver concentrations of PFASs in humans hampers our understanding of their hepatotoxicity and mechanisms of action. We investigated the partitioning of 11 PFASs between liver and blood in male CD-1 mice. Although accumulation of the perfluoroalkanesulfonic acids (PFSAs) in mice serum was higher than their carboxylic acids (PFCAs) counterparts as expected, the liver-blood partition coefficients ($R_{L/S}$) of PFSAs were lower than the PFCAs $R_{L/S}$, implying a competition between liver and blood. The *in vitro* experiments further indicated that the partitioning was dominantly determined by their competitive binding between human liver fatty acid binding protein (hL-FABP) and serum albumin (HSA). The binding affinities (K_d) of PFASs to both proteins were measured. The correlations between the $R_{L/S}$ and log $K_{d\ (hL\text{-}FABP)}$/log $K_{d\ (HSA)}$ were stronger than those with log $K_{d\ (hL\text{-}FABP)}$ alone, magnifying that the partitioning was dominantly controlled by competitive binding between hL-FABP and HSA. Therefore, the liver concentrations of the selected PFASs in humans could be predicted from the available serum concentrations, which is important for assessing their hepatotoxicity.



KEYWORDS: *PFASs, liver concentration, partition, protein binding, L-FABP, HSA*

INTRODUCTION

Per- and polyfluoroalkyl substances (PFASs) are widely applied in industrial and household products because of their favorable characteristics.[1] Due to the widespread production and persistence, PFASs have become ubiquitous environmental contaminants.[2] Perfluorooctanoate (PFOA) and perfluorooctanesulfonate (PFOS) are the most frequently detected PFASs in environmental compartments and are believed to exert potential toxic effects on human health.[3,4] Since 3 M Company stopped producing PFOS in 2000, the government agencies in many countries issued a series of regulations to restrict the production and application of PFOA and PFOS,[2,5] for example, PFOS was listed as an Annex B substance under the Stockholm Convention.[6] To meet the industrial demands, many novel PFASs were widely identified and applied as their alternatives and consequently released into the environment.[7] For example, perfluoropolyether carboxylic acids (PFECAs) as the alternatives of PFOA in polymerization processes, including hexafluoropropylene oxide dimer acid (HFPO-DA, and the commercial name is GenX), trimer acid, and tetramer acid (HFPO-TA, HFPO-TeA, respectively), were observed in river and coastal sea waters.[2,7,8] As alternatives of PFOS, which are applied as mist suppressant in hard chrome plating, 6:2 and 8:2 chlorinated polyfluoroether sulfonic acids (6:2 Cl-PFESA, 8:2 Cl-PFESA, and the commercial name is F-53B) were not only detected in the environmental media, including remote surface seawater,[9] but also observed in fish tissues[6] and human blood.[10] These two alternatives have longer half-lives in humans than PFOS. As a PFOS alternative, perfluoro-(2-ethoxyethane)-sulfonic acid (PFEESA) was occasionally detected (31%) in the nearshore seawater of China.[11]

It is well-documented that some PFASs may induce hepatotoxicity.[12−14] For example, PFOS caused significant hepatic steatosis and pathological micro/macro vacuolation with a superfluous triglyceride level in zebrafish larvae.[14] In mice, PFOS exposure dysregulated peroxisome proliferator-activated receptor γ (PPARγ) and then caused hepatic steatosis and disturbed hepatic lipid metabolism;[15] 6:2 and 8:2 Cl-PFESA displayed comparable or even higher agonistic activity toward the PPARs signaling pathway than PFOS in 3T3-L1 cells;[16] and the genes involved in the chemical carcinogenesis processes were strongly upregulated in liver due to HFPO-TA exposure.[13] These observations suggest that the liver is one of the target organs of these PFASs in organisms, where they have been shown to preferentially accumulate. However, histopathological

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studies to investigate the direct relationship between the accumulated levels of PFASs in liver and the induced hepatoxicities are rare, particularly in humans. Because it is difficult to abundantly sample internal human tissues (such as the liver) from living individuals, human blood is typically used to evaluate internal exposure to contaminants due to its easy accessibility and available analyses. Discovering the mechanisms underlying the partition of PFASs between organism livers and blood would assist to predict their concentrations and toxicities in livers.

As molecules enter organisms, they are absorbed into the bloodstream and transported to the organs within the body. Given the unique hydrophobic and lipophobic properties, accumulation of perfluoroalkanesulfonic acids (PFSAs) and perfluoroalkyl carboxylic acids (PFCAs) in organisms is driven not only by partitioning to phospholipids[17] but also predominantly by specific interactions with endogenous proteins.[18] Human serum albumin (HSA) is abundantly expressed in human blood, and the binding of chemicals with HSA plays critical roles in the pharmacokinetics, biodistribution, and toxicities of certain PFASs.[19] In human liver, liver-fatty acid binding protein (hL-FABP) is highly expressed and has strong binding affinities with long-chain PFASs for their transport, disposition, and elimination.[20] It was reported that the interaction of these long-chain PFASs with hL-FABP made an important contribution to the high bioaccumulation capacities of them in liver.[20,21] Thus, the accumulation of certain PFASs in liver is hypothesized to be the results of binding competition between the liver proteins and blood proteins.

Investigating the distribution of PFASs in different tissues provides hints to understand and speculate the intrinsic bioaccumulation mechanisms.[6] There are many field and laboratory studies reporting the tissue distribution of PFASs, but the results are quite diverse.[22] It seems that long-chain PFCAs (C_9−C_{14}) tended to be enriched in the rat liver; however, PFOA (C_8) and PFHpA (C_7) were preferably stored in the blood.[20] In contrast, HFPO-TA, as a novel PFOA alternative, preferred to accumulate in the liver rather than in serum.[13] While the highest total PFASs concentrations were observed in the kidney of frogs,[23] more than 75% of the whole body burden of F-53B was present in the blood, muscle, and gonad of crucian carp.[6] The underlying mechanisms accounting for the different partition behaviors of PFASs between liver and blood remain unclear. Particularly, there are very limited data regarding PFASs levels in human livers, which hampers the risk assessment related to the hepatoxicity of PFASs.

In the present study, we shed light on the partitioning mechanisms of 11 selected representative PFASs between human liver and blood. The main objectives are as follows: (i) elucidate the rule of liver-blood partition for diverse PFASs *in vivo*; (ii) simulate and disclose the mechanisms underlying the partitioning process using a self-developed *in vitro* experimental setup; and (iii) establish the liver-blood partition relationships between the binding affinities of PFASs with specific proteins in liver and serum of human. The findings help to address the dilemma for the growth concerns of potential hepatotoxicity while sparse and uncertain data of measured PFASs contents in human liver.

MATERIALS AND METHODS

Chemicals and Standards. Seven carboxylic acids (PFHxA, PFOA, PFNA, PFDoA, HFPO-DA, HFPO-TA, HFPO-TeA) and four sulfonic acids (PFOS, PFEESA, 6:2 Cl-PFESA, 8:2 Cl-PFESA) were included as target compounds. Due to the diverse functional groups of the target PFASs, carbon, oxygen, and sulfur atoms in the backbone were included when counting the chain length.[19] The detailed information about the used chemicals and reagents is listed in Text S1 and Table S1 of the Supporting Information (SI). A technical F-53B product, which contained 91% of 6:2 Cl-PFESA and 7% of 8:2 Cl-PFESA,[10] was separated using preparative high-performance liquid chromatography (HPLC) to obtain pure 6:2 Cl-PFESA and 8:2 Cl-PFESA (Text S1 and Table S2). Fatty acid-free albumin human serum albumin (HSA) was obtained from Sigma-Aldrich.

Recombinant Expression and Purification of hL-FABP. The expression and purification of hL-FABP were completed according to a previous study with slight modifications (Text S2).[24] The identification and purity (>95%) of the recombinant protein were confirmed by SDS-PAGE and Western blot (Figure S1).

Animal Treatment. Male CD-1 mice were used as the model animals, which have strong environmental adaptability, little individual difference, and lower elimination of PFASs than female to achieve higher body concentrations.[13,25] Specific pathogen-free (SPF) male CD-1 mice (aged 6 weeks, with a weight of 35 ± 3 g) were purchased from Beijing Vital River Laboratory Animal Technology Company (Beijing, China). Following 1 week of adaptation, they were randomly divided into four groups with each group consisting of six individuals (n = 6 per group) of equal size and acclimatized at standard condition (12 h light/dark cycle at 23 ± 1 °C, and humidity of 60 ± 5%). Three dosage levels (0.02, 0.2, and 2.0 mg/kg/day) were set accompanied by a blank control. The highest dosage was selected according to the no observed adverse effect level of PFOA in mice.[26] The mice received gavage dosage with mixed solutions (2.35, 23.5, and 235 μg/mL, respectively) of F-53B and the other nine PFASs at a rate of 8.5 mL/kg body weight (the mice were weighed before each gavage) every morning for 7 days. All the mice were fasted overnight and euthanized by cervical dislocation to sample liver and blood on the day 8 (Text S3). All of the animal procedures were approved (Text S1).

***In Vitro* Competition Experiments.** *In vitro* experiments were designed to investigate the competitive binding of the target PFASs with proteins in blood and liver,[27,28] which were individually housed in two dialysis tubes, and the setup is shown in Figure S2. The dialysis tubes made of biotech grade cellulose ester (CE) membrane (Float-A-Lyzer G2, 3.5−5 kDa) were obtained from Spectrum Laboratories, Inc. Two tubes (serum or HSA in tube S, while liver lysate or hL-FABP in tube L) were placed in a polypropylene beaker containing Tris-HCl buffer at 25 °C. Preliminary experiments proved that there was negligible mass loss of PFASs due to adsorption to the membrane and inner wall of the beaker, indicating that the competition experiments were reliable (Text S4 and Figure S3). Two groups of experiments were designed: in Group I, the liver lysate and serum collected from the same mouse were housed, respectively, in the two tubes to explore the competitive binds of PFASs between them; while in Group II, the proteins in human liver (hL-FABP) and serum (HSA) were put in the tubes to investigate the competitive binding between them, and the protein concentrations were selected according to their levels in the mice body (Text S4). Three levels of PFASs were set for each group, which were in line with the *in vivo* exposure experiment. The PFASs bound with the liver lysate or serum and the proteins are calculated according to the following equations:

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$$[\mathrm{PFASs}]_{\mathrm{L\ or\ S}} = [\mathrm{PFASs}]_{\mathrm{tube\ L\ or\ tube\ S}} - [\mathrm{PFASs}]_{\mathrm{free}} \quad (1)$$

$$[\mathrm{PFASs}]_{\mathrm{hL\text{-}FABP\ or\ HSA}} = [\mathrm{PFASs}]_{\mathrm{tube\ L\ or\ tube\ S}} - [\mathrm{PFASs}]_{\mathrm{free}} \quad (2)$$

where $[\mathrm{PFASs}]_{\mathrm{tube\ L\ or\ tube\ S}}$ is the concentration of the PFASs in tube L or tube S and $[\mathrm{PFASs}]_{\mathrm{free}}$ refers to the concentration of PFASs in the Tris-HCl buffer. $[\mathrm{PFASs}]_{\mathrm{L\ or\ S}}$ (ng/mg) represents the mass of PFASs (ng), which binds to one milligram of liver or serum. $[\mathrm{PFASs}]_{\mathrm{hL\text{-}FABP\ or\ HSA}}$ (ng/mmol) is the mass of PFASs (ng) that binds to one millimole of hL-FABP or HSA.

Extraction and Instrumental Analysis. PFASs extraction from the mice tissues, serum, and liver lysate in the equilibrium dialysis were performed using an ion-pairing method followed by a solid phase extraction (SPE) cleanup as previously described (Text S3).[13] In equilibrium dialysis, the target PFASs in Tris-HCl buffer were purified and enriched using Cleanert PEP cartridges, while precipitated buffer was used to extract the PFASs from the hL-FABP and HSA solutions (Text S5).[19] The targeted PFASs in all the samples were quantified on an ultraperformance liquid chromatography system (UPLC) coupled with a Waters Xevo TQ-S tandem mass spectrometer (MS/MS, Waters, Milford, MA) operated with negative electrospray ionization (ESI) in multiple reaction monitoring (MRM) mode. Detailed instrumental parameters and analytical methods are supplied in Text S6 and Table S3.

Quality Assurance and Quality Control. The determination of the 11 PFASs was based on the EPA methods 537.1[29] and 8237.[30] A procedural blank and a solvent blank were analyzed with each batch of 10 samples to monitor any background contamination and instrument carryover. The method detection limits (MDLs) were determined as the average concentrations in the blanks plus three times the standard deviations when specific PFASs were detected in the procedural blanks. Otherwise, the MDLs were determined as the concentrations producing a signal-to-noise ratio of three (Table S4). The blank sample matrixes including the mouse liver, serum, and liver lysate were prepared exactly the same as the samples (Text S3), and the PFASs concentrations in the samples were quantified using matrix-matched calibration standards (0–60 ng/mL). The PFASs in the Tris-HCl buffer and HSA/hL-FABP solutions in the equilibrium dialysis were quantified by calibration standards in methanol with similar concentration ranges. Quantitation was carried out using the internal standard method for the target PFASs in all samples (Table S3). The recoveries in the spiked control samples were in the range 70–120% (Table S4 and Table S5), and the PFASs concentrations were calculated without recovery correction.

Determination of Binding Capacity. Microscale thermophoresis (MST) is a promising technique to quantify ligand–protein interactions, which has the advantages of using less proteins and short reaction time. The dissociation constants (K_d) values were measured by MST using Monolith NT.115 (NanoTemper Technologies, Munich, Germany) following the manufacturer's instructions (Text S7). Thermodynamic parameters were analyzed to understand the interaction models, and they were fitted by the Van't Hoff equation (eq 3)[31]

$$R \ln\left(\frac{K_d}{C_0}\right) = -\frac{\Delta H}{T} + \Delta S \quad (3)$$

where C_0 is the standard state concentration (1 mol/L), R is the ideal gas constant (8.314 J·mol^{-1}·K^{-1}), and T is the absolute temperature.

Molecular Docking Analysis. The intermolecular interactions were characterized by molecular docking analysis.[32] The optimized configurations of all the 11 PFASs were obtained at the B3LYP/6-31G(d) level by the Gaussian 16 program package. Since the secondary binding site of hL-FABP made negligible contribution,[20,33] we only considered the primary binding site in the inner pocket. The crystal structure of hL-FABP, containing the complete key residues with high complex resolution, was downloaded from the Protein Data Bank (PDB code: 3STM; htpp://www.rcsb.org/pdb). Both protein and ligand structures were pretreated with AutoDock Tools (v1.5.6),[34] and the output files were used for docking with Autodock Vina (v1.1.2).[35] The protein–ligand complex was illustrated and analyzed by PyMOL (v2.4).[36] The reliability of the docking results were verified by palmitic acid (PLA), which was successfully docked in compliance with the crystallized complexes with an acceptable root-mean-square deviation (RMSD, 1.94 Å).

Extrapolation $R_{L/S}$ from Mouse to Human. Taking into consideration the interspecies variation between mouse and human physiological parameters, the mice $R_{L/S}$ were adjusted to humans using the ratio VLC/VPlasC, where VLC represents the fractional volume of liver in body and VPlasC represents the fractional volume of plasma in body. Their values for both human and mouse were obtained from a previous report.[37]

Statistical Analysis. Data analyses were carried out by Graphpad Prism 9 (Graphpad Software Inc.). One-way analysis of variance (ANOVA) with a Tukey test was used to estimate the significant differences between different groups. Spearman's rank correlations analysis was conducted to explore the PFASs partition in mice liver and serum, and the competitive binding in equilibrium dialysis experiments. Statistical significance was defined as $p < 0.05$ for all comparisons.

RESULTS AND DISCUSSION

Liver and Body Weight of the Mice. For the 0.02 and 0.2 mg/kg/d dosage groups, the liver and body weights of the mice did not change distinctly after 7 days of exposure (Figure S4a). While the absolute liver weight in the highest dosage group increased distinctly, the percentage of liver in the body weight (13.3%) was significantly higher than the other groups (Figure S4b). The increased liver weight might be due to some adverse effects according to previous toxicity studies.[13,38]

Partitioning of PFASs in Mice Serum and Liver. In all the three exposure groups, the PFASs with medium- and long-chain lengths, such as PFOA, PFNA, PFDoA, HFPO-TA, HFPO-TeA, PFOS, and 6:2 and 8:2 Cl-PFESA, were prone to partition into the liver rather than in the serum (Figure S5). For the short-chain PFASs, such as PFHxA, HFPO-DA, and PFEESA, their concentrations were significantly lower in both liver and serum, suggesting that they had very low accumulation potentials in liver and serum. For the ordinary PFCAs, their liver concentrations increased from PFHxA (C_6) to PFDoA (C_{12}) in all dosage groups, which was similar to the results reported previously.[20] Different from the liver, the serum concentrations of PFCAs decreased from PFOA (C_8) to PFDoA (C_{12}). For HFPO analogues, the liver concentration of HFPO-TeA was slightly lower than HFPO-TA in the lowest dosage group, while that of HFPO-TeA was 3.26- and 3.15-fold higher than HFPO-TA in the medium and high dosage groups, respectively (Figure S5b,c).

The different variation trends of PFASs in serum and liver suggested that they had different binding affinities and

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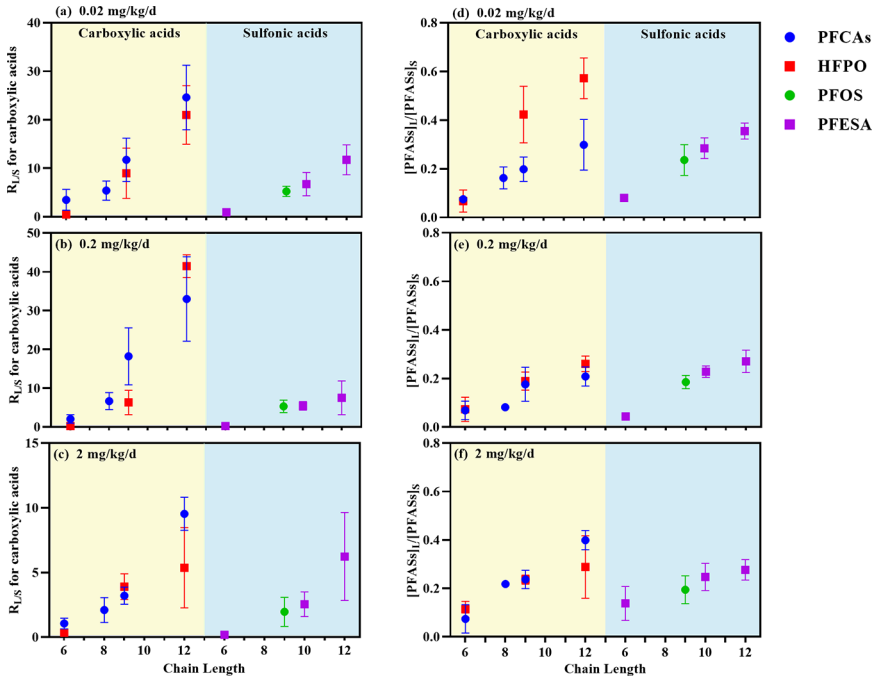


Figure 1. (a–c) $R_{L/S}$ and (d–f) $[PFASs]_L/[PFASs]_S$ vary with chain length of PFASs in (a and d) 0.02 mg/kg/d, (b and e) 0.2 mg/kg/d, and (c and f) 2 mg/kg/d dosage groups. $R_{L/S}$ is the ratio of concentration for PFASs in liver to serum in CD-1 mice, and $[PFASs]_L/[PFASs]_S$ is the ratio of concentration for PFASs bound with liver to that with serum in equilibrium dialysis experiments. The chain length was the counts of total atoms in backbone of PFASs (for PFCAs, the number of carbon atoms; for HFPO analogues, the number of carbon and oxygen atoms; for PFOS, the number of carbon and sulfur atoms, and for PFESA, the number of carbon, oxygen, and sulfur atoms in backbone).

mechanisms with the proteins in both tissues. The liver-blood partition coefficient $R_{L/S}$, which is the concentration ratio in liver to serum, was calculated to indicate the partitioning of the target PFASs between liver and serum (Figure 1). The chain length and oil–water partition coefficient (log *P*, predicted by Advanced Chemistry Development Software V11.02 in Table S6) have been demonstrated as important indices for the bioaccumulation of PFASs in organisms.[18] Strong positive correlations were observed between $R_{L/S}$ and the chain length as well as log *P* (r_s = 0.71 and 0.63, respectively). This suggested that the tissue partition of the individual PFASs was greatly impacted by their specific molecular properties.[39]

The relatively low $R_{L/S}$ values of several compounds with small molecular size and low log *P*, such as PFHxA (1.1–3.5) and PFEESA (<1), implied that they are inclined to transport elsewhere in the body by bloodstream rather than store in liver.[20] On the contrary, 6:2 and 8:2 Cl-PFESAs with higher $R_{L/S}$ values and lower elimination rates[10] than PFOS were inclined to accumulate in liver to cause potential hepatotoxicity. Similar to other mammals, the $R_{L/S}$ values of the PFCAs, HFPO analogues and PFESAs exhibited a positive monotonous relationship with carbon chain length and were higher than 1 at all dosage levels. However, the $R_{L/S}$ values of PFOA, PFNA, and PFDoA in fish and seabirds were generally less than 1 (Figure S6). This suggested that these compounds behaved differently in fish and mammals, and they tended to accumulate in liver relative to serum in mammals. It is worth mentioning that all the target PFASs exhibited lower $R_{L/S}$ values (0.2–9.5) in the highest dosage group than the other two groups. Similar results were reported for PFOA and HFPO-TA in male BALB/c mice (Figure S6).[13] Consistent with many previous studies, the PFASs with sulfonic acid functional groups had higher bioaccumulation potentials in liver and serum than their carboxylic counterparts;[40] however, the $R_{L/S}$ values of the PFASs with a sulfonic acid group were generally lower at all dosage levels. This discrepancy suggested that there was competition for binding with these PFASs between liver and blood.

***In Vitro* Competition Experiments.** To discover the intrinsic mechanisms for the observed *in vivo* results in mice, *in vitro* competition experiments were setup on the basis of equilibrium dialysis, which has been widely used to measure the binding affinities of drug compounds with proteins.[41,42] In Group I, $[PFASs]_L/[PFASs]_S$ (which represents the ratio of the bound PFASs, the same below) increased with the chain length (Figure 1), and there were significantly positive correlations between *in vitro* $[PFASs]_L/[PFASs]_S$ and *in vivo* $R_{L/S}$ at all three levels (r_s = 0.80, 0.70, and 0.92, respectively, from low to high levels of PFASs, Figure S7). The results suggested that the tissue-specific distribution of PFASs in the mammal's liver and serum was due to the partition of PFASs between the proteins in the liver lysates and serum.[18,21] To further understand the partition between liver and blood in human, Group II was designed employing the highly expressed intracellular proteins in human liver and blood, hL-FABP, and HSA, which were reported to account for PFASs bioaccumulation in the two tissues.[21] As expected, $[PFASs]_{hL\text{-}FABP}/[PFASs]_{HSA}$ increased with the chain length (Figure S8), and stronger correlations between $R_{L/S}$ and $[PFASs]_{hL\text{-}FABP}/[PFASs]_{HSA}$ were observed, with r_s = 0.97, 0.90, and 0.89 for the three dosage groups, respectively (Figure 2). These results affirmed that the competitive binding of PFASs with the two proteins played a key role in the liver-blood partitioning of the target PFASs in humans.

Interactions of PFASs with hL-FABP and HSA. Although the binding affinities of some common PFASs with hL-FABP

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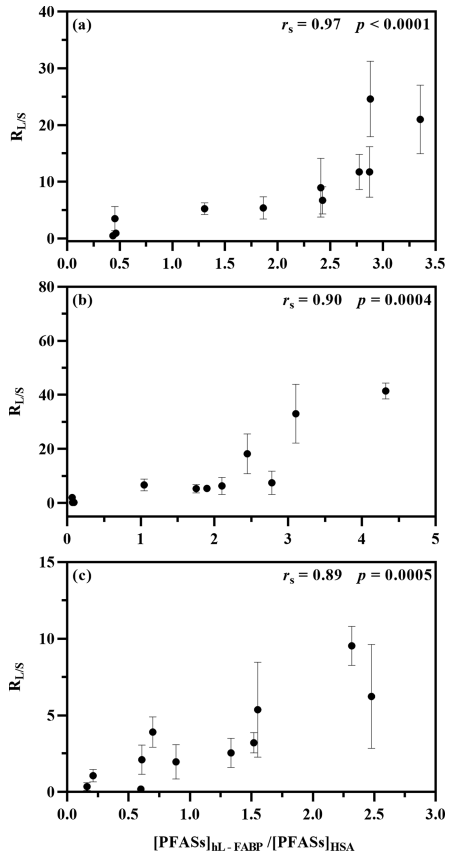


Figure 2. Spearman correlation between [PFASs] $_{hL-FABP}$/[PFASs]$_{HSA}$ and $R_{L/S}$ in (a) 0.02 mg/kg/d dosage group, (b) 0.2 mg/kg/d dosage group, and (c) 2 mg/kg/d dosage group.

and HSA were reported previously, they are not available for the novel PFASs alternatives. Particularly, the intrinsic mechanisms of the binding process for these alternatives remain unclear; therefore, MST was applied to characterize the interactions between them (Figures S9 and S10). The $K_{d\ (hL-FABP)}$ values (Table 1 and Table S7) of the PFASs measured at different temperatures (from 21 to 40 °C) decreased with chain length, suggesting that their binding affinities were stronger for long-chain PFASs. The negative Gibbs free energy (ΔG) for all the target PFASs (Figure S11) indicated that the binding process was spontaneous. The negative enthalpy change (ΔH) and standard entropy change (ΔS) suggested that hydrogen bond and van der Waals forces were the main drivers of the binding process.[43] Similar to PLA, molecular docking results indicated that there was hydrogen bonding between the charged headgroups of PFASs and the Arg, Ser, or Thr residues of hL-FABP (Figure S12). The hydrophobic tails stretch to interact with nonpolar residues, leading to stronger binding for the more hydrophobic (CF_2) units of PFASs. Specially, PFDoA with a much longer carbon length has to bend to fit the pocket.[44] HFPO homologues and PFESA, which contain oxygen atoms in the backbone, would distort to a certain extent in the cavity.[13,33] For these compounds, electrostatic interaction was more relevant to $K_{d\ (hL-FABP)}$ than van der Waals interaction (r = −0.94 vs −0.63).[20] This suggested that the insertion of oxygen atoms in the backbone may change the binding modes of PFASs with hL-FABP.[45]

Regarding the binding with HSA, stronger binding affinities were observed for the medium-chain length analogues (e.g., HFPO-TA > HFPO-TeA > HFPO-DA), which was similar to the trend of PFASs concentrations in the serum (Table 1). The larger size of the sulfonate moiety induced greater hydrophobic and electrostatic interaction for sulfonic acids than carboxylic acids with a similar chain length to HSA (e.g., PFOS > PFNA and HFPO-TA).[46]

Associations between Liver-Blood Partition and Binding Affinities. Up to now, there is sparse information on the interaction of the target PFASs with mouse L-FABP and serum albumin. Since mouse L-FABP (84.3%)[47] and serum albumin (72.4%)[48] have high identities with humans, we investigated the relationships between $R_{L/S}$ obtained from mice and log $K_{d\ (hL-FABP)}$/log $K_{d\ (HSA)}$, to further verify the competitive binding mechanisms in liver-blood partition. As expected, there were strong negative correlations between the $R_{L/S}$ and log $K_{d\ (hL-FABP)}$/log $K_{d\ (HSA)}$ (Figure 3, Table S8, and Correlations 1 and 2). The linear regressions for the low and medium dosage groups were similar; thus, they were grouped together, which exhibited a steeper slope (−25.60) than that generated from the highest dosage group (−6.18). The plausible reason is that there was a fixed number of binding sites in liver, which would be saturated at higher levels of PFASs.[13] A previous study reported the inverse relationships between bioaccumulation factors in organisms and PFASs exposure concentrations, which was believed to be related to the limited number of protein binding sites.[49]

The binding affinities of PFASs to hL-FABP were significantly correlated with the human placental transfer efficiencies[19] and rat liver concentrations.[20] We thus examined the association between $R_{L/S}$ with log $K_{d\ (hL-FABP)}$ (Figure S13, Table S8, and

Table 1. Binding Affinities of PFASs with hL-FABP at 25 °C and Dissociation Constants of PFASs with HSA

	hL-FABP				HSA
PFASs	K_d (μM)	ΔS (J/mol)	ΔH (kJ/mol)	ΔG (kJ/mol)	K_d (μM)
PFHxA	538.07	−58.16	−35.96	−18.62	195[19]
PFOA	17.95	−46.70	−41.31	−27.39	115[19]
PFNA	6.62	−44.30	−42.87	−29.66	73[19]
PFDoA	3.45	−196.20	−89.15	−30.65	103[19]
HFPO-DA	267.78	−31.50	−29.97	−20.57	1156
HFPO-TA	19.94	−16.12	−31.69	−26.89	84
HFPO-TeA	1.22	−73.09	−55.34	−33.55	179
PFOS	16.45	−31.83	−37.02	−27.53	38[19]
PFEESA	3138.20				874
6:2 Cl-PFESA	12.47	−259.20	−105.07	−27.79	67[19]
8:2 Cl-PFESA	5.39	−62.10	−48.57	−30.05	31

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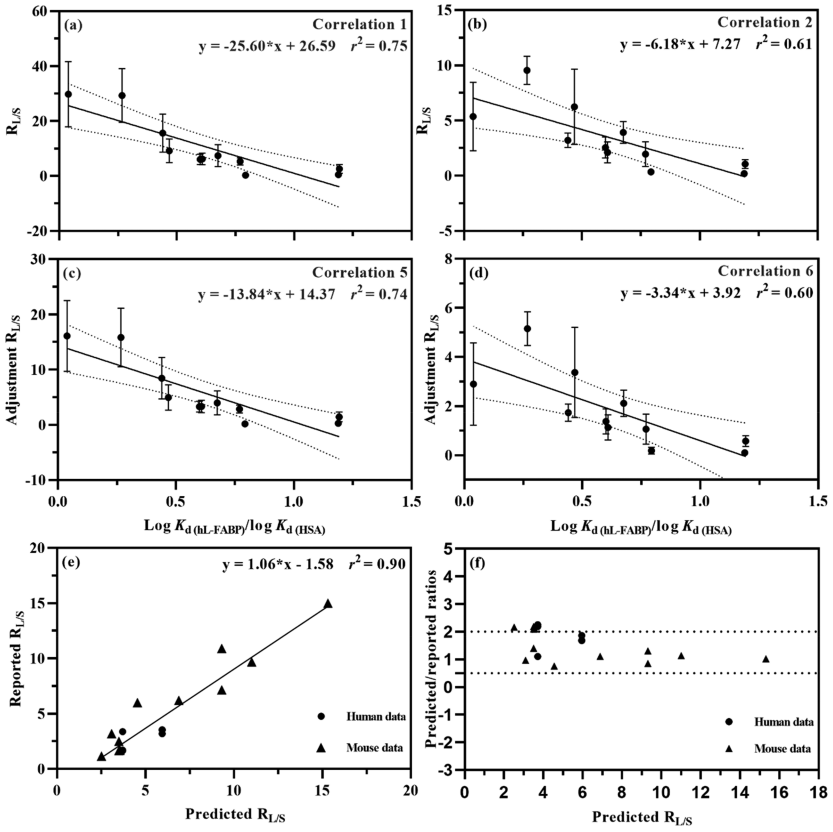


Figure 3. Correlation between log $K_{d\ (hL\text{-}FABP)}$/log $K_{d\ (HSA)}$ with (a) mice $R_{L/S}$ and (c) humans $R_{L/S}$ for low and medium dose exposure groups (Correlations 1 and 4); and (b) mice $R_{L/S}$ and (d) humans $R_{L/S}$ for high dose exposure group (Correlations 2 and 5). The dotted line represents the 95% confidence interval. (e) Goodness-of-fit curve between predicted and reported data of $R_{L/S}$ for mice and humans. (f) Predicted to reported ratios of $R_{L/S}$. The reported $R_{L/S}$ are available in Table S9.

Correlations 3 and 4), for which the correlation was weaker than that of $R_{L/S}$ and log $K_{d\ (hL\text{-}FABP)}$/log $K_{d\ (HSA)}$. This fortified that the bioaccumulation of PFASs in liver was not only determined by their binding affinities with hL-FABP alone but also more controlled by their competitive binding between hL-FABP and HSA.

The established correlations were extrapolated to humans using the adjusted $R_{L/S}$ (Figure 3, Table S8, and Correlations 5 and 6). The tolerable weekly intake of PFOA (6 ng/kg/week) suggested by European Food Safety Authority[50] is 3 orders of magnitude lower than the highest dosage level, which was corresponding to a human equivalent dose (HED, 37.1 μg/kg/week).[26] Therefore, the Correlation 5 is generally applicable for the general population, while Correlation 6 is more suitable for occupational population.

On the basis of the established correlations, the $R_{L/S}$ values of the 11 PFASs in mice (Correlations 1 and 2) and general human populations (Correlation 5) were predicted. They were compared with the results obtained from previous related studies to verify the universality of the developed correlations (Table S9). As shown in Figure 3e,f, the predicted $R_{L/S}$ values were in agreement with the measured results (slope of fitting curve was 1.06, with r^2 = 0.90), and the predicted-to-reported ratios for almost of all the points were within a factor of 2, meeting the WHO model precision criteria.[51] On the basis of the predicted $R_{L/S}$ values of PFASs in human, their human liver concentrations could be estimated on the basis of the measured serum concentration. For example, according to Correlation 5, the $R_{L/S}$ of PFOA in the general population is 5.94. It was reported that the mean plasma concentration of PFOA was 3.2 ng/mL in Spain, which was equal to a serum concentration of 3.84 ng/mL.[52,53] Thus, the liver PFOA concentration was estimated to be 22.57 ng/g, which was within the concentration range in the Spanish liver donors.[53] The good performance of the established correlations promoted us to further calculate the concentration of protein-bound PFASs in liver ($[PFAS]_{hL\text{-}FABP}$) (Text S8). For example, the plasma concentration of PFOS in the general German population was 1.8 ng/mL;[54] thus, the serum concentration was 1.98 ng/mL (the PFOS concentration ratio in serum to plasma is 1.1[52]). The $R_{L/S}$ of PFOS in the general population predicted by Correlation 5 is 3.72. The total PFOS concentration in liver was estimated to be 7.36 ng/g while that bound with hL-FABP was 6.36 ng/g. This suggested that a large fraction of PFOS might be delivered to nuclear receptors by hL-FABP and thus result in the potential hepatotoxicity.

IMPLICATIONS

In this study, we revealed that the competitive binding of certain PFASs with hL-FABP and HSA in humans dominantly determined its partitioning between liver and blood. Significant correlations between $R_{L/S}$ and log $K_{d\ (hL\text{-}FABP)}$/log $K_{d\ (HSA)}$ allow us to predict the human liver concentrations of the target PFASs on the basis of the available human serum concentrations, which resolved the scare data conundrum for measured PFASs concentrations in human liver. Thus, the health status of human liver could be evaluated by coupling with other physiological parameters. However, interspecies variations may cause some uncertainty and variability in the results. In

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addition, some membrane transporters, which were believed to be involved in the metabolic process for PFASs,[55] such as enterohepatic circulation, were not considered here. More studies are warranted to understand the contribution of other proteins to the partitioning of PFASs among target tissues to produce more accurate toxicokinetic research and PFASs health assessments.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.1c08493.

Discussions of chemicals and animals used, recombinant expression and purification of hL-FABP, pretreatment procedures of serum and liver samples, equilibrium dialysis experiments, pretreatment procedures of Tris-HCl buffer, HSA, and hL-FABP solution, instrumental parameters and analytical methods, determination of binding capacity, and calculations of the protein-bound PFASs concentration in liver, tables of detailed information of 11 PFASs in the research, instrumental parameters and separation method to prepare pure 6:2 Cl-PFESA and 8:2 Cl-PFESA from F-53B, target analytes and internal standards used in the present study, recoveries and method detection limits for 11 PFASs, predicated physicochemical properties and parameters of PFASs, dissociation constants of PFASs to hL-FABP at different temperatures, significance, residuals, Y-intercept, standard errors, and 95% confidence intervals for the fitting curve, and data of the reported $R_{L/S}$ for humans and mice, and figures of SDS-PAGE and Western blot, simulation diagram of equilibrium dialysis experiments, fold change of the PFAS concentration in terminal buffer to the initial buffer in the adsorption tests, liver and body weight, ratio of liver to body weight of mice in four PFASs dosage groups, concentration of PFASs in mice liver and serum, comparison of the $R_{L/S}$ values of PFASs for different species in previous reports, Spearman correlation, interaction of PFASs with hL-FABP and HSA measured by microscale thermophoresis, thermomechanical analysis of the interaction between PFASs with hL-FABP by fitting K_d at different temperatures, proposed binding conformations of the complexes of PFASs and PLA to hL-FABP, and relationship between $R_{L/S}$ with log $K_{d\ (hL\text{-}FABP)}$ for low-dose exposure groups and high-dose exposure groups (PDF)

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Author Contributions

‖Y.J. and Y.Z. contributed equally to this paper.

Notes

The authors declare no competing financial interest.

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